AI-Driven Interactive Wine Cellar Management System with Transparent Display Integration

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***Abstract*—The premium home appliance sector is increasingly prioritizing artificial intelligence (AI) and user experience (UX) over mere hardware specifications. In the context of luxury wine preservation, existing commercial cellars remain passive climate-control units, while third-party software solutions suffer from high interaction friction and a lack of physical-digital synchronization. To address these bottlenecks, this paper presents an AI-driven, transparent-display-integrated wine cellar management system designed for high-end collections. Our system introduces a robust Vision AI pipeline for automated label recognition and spatial mapping, effectively eliminating manual inventory tracking even for flexibly stacked bottles. Furthermore, we implement a privacy-preserving, on-device Multi-Agent Small Language Model (SLM) architecture. By leveraging a quantized 4-bit base model with dynamic Low-Rank Adaptation (LoRA) swapping on a low-power Neural Processing Unit (NPU), the system generates persona-driven expert dialogues, dynamic food pairings, and personalized daily briefings without violating the stringent thermal and memory constraints of the appliance. Ultimately, this architecture transforms the traditional wine cellar from a passive storage unit into an actively managed, hyper-personalized asset management ecosystem.**

***Index Terms*—Smart Home Appliance, Edge AI, Vision AI, Small Language Model (SLM), Low-Rank Adaptation (LoRA), Wine Management System, Human-Computer Interaction (HCI)**

Roles	Name	Task description and etc.
User	Chiang Jun Han	Describe typical user scenarios and workflows. Identify usability problems and practical needs. Try prototypes and provide feedback.
Customer	Naomi K. Fossaberg	Define the project's purpose and desired outcomes. Prioritize requirements and features. Evaluate whether the final solution meets the agreed requirements.
Software developer	Enkhbold Myagmarjav	Design and implement software components. Write and review code. Test functionality and fix bugs.
Development manager	Sungbin Bae	Plan tasks, milestones, and deadlines. Coordinate collaboration and monitor progress. Identify risks and help resolve blockers.

Table 1. Role Assignments

I. INTRODUCTION

A. Motivation

In recent years, the premium home appliance market has undergone a fundamental paradigm shift, transitioning from pure hardware performance competition toward elevating user experience (UX) through Cloud-based Artificial Intelligence (AI) and the seamless connectivity of the Internet of Things (IoT). High-end luxury product lines, such as LG SIGNATURE, incur substantial hardware manufacturing costs, thereby demanding sophisticated, cloud-backed software capabilities that deliver commensurate value to justify their premium positioning. Concurrently, the global expansion of the fine wine market has driven rapid demand for specialized home wine cellars driven by AI management. However, existing commercial wine cellars remain passive, isolated appliances limited to fundamental condition control (temperature, humidity regulation, anti-vibration, and so on), failing to leverage vast

cloud computing resources to satisfy the sophisticated digital management needs of modern fine wine collectors.

B. Problem Statement and Target Audience

The primary target demographic for luxury wine cellars consists of high-net-worth individuals, predominantly in their 40s to 60s. Although these collectors frequently possess extensive and valuable wine collections, current management paradigms introduce critical usability bottlenecks:

1) **High Interaction Cost and Friction:** Existing mobile-app-based wine management solutions force users to repeatedly handle cold bottles while operating small smartphone screens. This physical separation between the appliance and the management interface induces high cognitive load and interaction friction.
2) **Absence of Physical-Digital Synchronization:** Current wine cellars lack real-time synchronization between physical storage and digital databases. Consequently, users are required to perform manual inputs for every bottle movement, leading to severe tracking discrepancies and user abandonment over time.

C. Related Software and Limitations

An analysis of existing wine management software (e.g., Vivino, CellarTracker) and proprietary cellar tracking apps reveals distinct functional limitations when applied to high-end collections:

- **Absence of Spatial Awareness:** Most commercial applications function merely as flat digital ledgers or list-based inventories. They lack a robust spatial mapping mechanism that correlates a specific bottle's data with its exact physical coordinates (e.g., specific racks and rows) within the cellar, inevitably forcing users to manually search through shelves.
- **Static Data without Environmental Context:** Existing software operates completely independently of the cellar's internal hardware sensors. Consequently, they cannot factor real-time environmental data—such as multi-zone temperature differences—into their aging algorithms or serving readiness calculations.
- **Reactive and Monotone User Experience:** Current tools are inherently reactive, requiring users to explicitly query the database to check a wine's optimal drinking window or food pairing suitability. Furthermore, they rely on static text reviews, lacking the proactive intelligence and dynamic, expert-level engagement necessary to autonomously curate an immersive experience.

D. Proposed Solution and Key Contributions

To overcome these limitations, this paper proposes a cloud-powered, AI-driven wine cellar management system integrated with a transparent display door. Designed specifically for luxury appliances such as the LG SIGNATURE line, the proposed architecture leverages the robust computing infrastructure of centralized datacenters (e.g., LG Datacenter) to deliver uncompromising performance.

The key contributions of this work are summarized as follows:

- **Vision AI and Automated Spatial Mapping:** Integrates a robust vision pipeline to extract label and vintage data. Upon insertion, an automated placement mechanism registers the bottle's location. This extends beyond rigid 2D grids, possessing the capability to accurately map bottles even when they are flexibly stacked on top of one another without discrete rack partitions.
- **Cloud-Backed Multi-Agent LLM Architecture:** Replaces resource-constrained edge computing with a high-performance cloud backend (LG Datacenter). This paradigm shift allows the system to deploy massive, state-of-the-art Large Language Models (LLMs) without imposing any thermal, processing, or memory constraints on the cellar appliance itself. The cloud architecture seamlessly orchestrates multi-agent dialogues among critic personas, ensuring hyper-personalized, expert-level interactive experiences.
- **AI-Driven UX Enhancement:** Transcending basic touch interactions, the cloud-integrated AI engine drastically elevates the overall user experience. By analyzing multidimensional real-time data, the system proactively offers a broad spectrum of personalized management features—such as optimal drinking window alerts and dynamic food pairings—transforming passive wine storage into an actively managed ecosystem.
- **Multi-Device Distributed Scalability:** Establishes a synchronized system architecture that seamlessly unifies multiple physical cellar units under a single user profile. Leveraging cloud centralization, it guarantees real-time synchronization between the mobile application and the appliance, eradicating tracking discrepancies.

II. REQUIREMENTS SPECIFICATION

A. Functional Requirements

1) User Management:

- The system shall allow a user to create an account with the following mandatory information: Username, Email, and Password.
- The system shall allow a registered user to log in and out.
- The system shall allow an authenticated user to edit their profile information.

2) Inventory Mapping:

- The system shall automatically detect wine bottles placed inside the wine cellar using computer vision.
- The system shall extract wine label information from detected bottles using AI.
- The system shall automatically register newly detected bottles in the inventory.
- The system shall automatically associate each detected bottle with a physical location in the wine cellar.
- The system shall automatically update a bottle's location when it is moved within the wine cellar.

- The system shall assign a unique identifier to each physical wine bottle to prevent duplicate inventory records.
- The system shall allow an authenticated user to review and correct AI-extracted wine information before confirming registration.
- The system shall automatically detect stacked wine bottles.
- The system shall automatically remove a bottle from the inventory when it is removed from the wine cellar.
- The system shall allow an authenticated user to manually add a wine bottle to the inventory.
- The system shall allow an authenticated user to manually modify bottle information.
- The system shall allow an authenticated user to manually remove a wine bottle from the inventory.

3) Inventory Search & Visualisation:

- The system shall allow a user to search for a bottle by wine name, winery, region, vintage or grape variety.
- The system shall display the physical location of a selected bottle within the wine cellar.
- The system shall display a visual map of bottles stored in the wine cellar.

4) Environmental Monitoring:

- The system shall collect environmental data from the wine cellar sensors.
- The system shall store historic environmental measurements from the wine cellar.
- The system shall associate environmental conditions with individual bottle locations.
- The system shall notify users when temperature or humidity exceeds user-defined thresholds for a specified duration.
- The system shall notify users when sensors become disconnected or environmental data is outdated.

5) AI Recommendations:

- The system shall allow a user to input data regarding a meal and receive wine pairing recommendations from the user's current inventory.
- The system shall allow a user to select a specific wine from the inventory and generate complementary food pairing suggestions, including culinary recipes or food delivery menu categories.
- The system shall recommend bottles that have reached their optimal drinking window.
- The system shall notify a user when wines are approaching their optimal drinking window.
- The system shall generate personalised recommendations based on historical consumption patterns.
- The system shall allow a user to input specific constraints, such as a target budget, flavor profile, or event theme, to request a curated wine list.
- The system shall automatically generate and display a customized wine list that satisfies the user-defined parameters, utilizing both the existing cellar inventory and external databases for potential new purchases.
- The system shall recommend suitable storage locations for newly added bottles.
- The system shall recommend wines via natural language queries processed by the cloud-based LLM.

6) Personalized Daily Wine Briefing:

- The system shall aggregate daily global wine news, vintage reports, and market trends from external web sources and APIs.
- The system shall personalize the aggregated news content by cross-referencing it with the user's specific cellar inventory, highlighting articles relevant to their owned regions, grapes, or producers.
- The system shall automatically curate and display a "Daily Wine Briefing" on the cellar's front display, including inventory-specific insights (e.g., wines reaching their peak, humidity stability).
- The system shall send a summarized version of the daily briefing as a push notification to the user's synchronized mobile application at a user-defined time.

7) Real-Time Market Value Integration:

- The system shall integrate with external global wine market APIs (e.g., Wine-Searcher) to retrieve real-time pricing data for the specific vintages stored in the cellar.
- The system shall calculate and visualize the real-time aggregated market value of the user's entire wine collection on the front display.
- The system shall provide graphical visualizations of historical price trends and value appreciation for individual bottles to assist the user in asset management.

8) Multi-Cellar Management:

- The system shall allow a user to manage multiple wine cellar units and wine cellars under a single account.
- The system shall synchronise inventory information across connected wine cellars.
- The system shall allow a user to search across all connected wine cellars.

9) Cloud-Powered Multi-Agent LLM Dialog System:

- The system shall integrate a cloud-hosted Multi-Agent Large Language Model (LLM) framework, configured with multiple distinct personas reflecting the evaluation tendencies and grading philosophies of renowned wine critics (e.g., Robert Parker, Luca Maroni).
- The system shall automatically send an API request to the cloud backend to trigger a simulated multi-agent dialogue whenever a new wine is registered to the cellar inventory or actively selected by the user via the touch display.
- The system shall retrieve and visualize interactive, chat-style debates on the front display in real-time, showcasing diverse analytical perspectives, tasting notes, and aging predictions for the selected wine.
- The system shall allow the cloud-based AI to continuously fine-tune its conversational outputs and recommendations by securely analyzing the user's synchronized cellar database, including inventory history, consumption patterns, and individual preferences.

- **Cloud Base Model Operation:** The system shall utilize a centralized, high-parameter Large Language Model (LLM) hosted on the data center as the foundational reasoning engine, removing the need for aggressive quantization.
- **Server-Side Dynamic LoRA Swapping:** The system shall dynamically load and unload specific Low-Rank Adaptation (LoRA) weight matrices corresponding to the active persona on the cloud server during multi-agent dialogue generation, maximizing GPU VRAM efficiency while ensuring perfect persona emulation.
- **Continuous Persona Expansion:** The system shall allow the seamless addition of new critic or user-specific personas entirely via server-side LoRA updates, requiring no local software patches or model downloads on the cellar appliance.

10) Mobile Application Integration:

- The system shall provide a companion mobile application that synchronizes real-time inventory and cellar status with the user's account.
- The system shall allow a user to remotely browse their wine inventory, search for bottles, and view AI recommendations via the mobile application.
- The system shall push real-time notifications to the mobile application regarding critical environmental alerts (e.g., temperature fluctuations) and optimal drinking windows.
- The system shall allow a user to remotely adjust the temperature and humidity settings of individual cellar zones through the mobile application.
- The system shall allow a user to manually add, edit, or remove wine inventory data via the mobile application when away from the physical cellar.

B. Non-Functional Requirements

1) Performance:

- The system shall return inventory search results within 2 seconds for 95% of requests.
- The system shall update the displayed inventory within 10 seconds after a bottle had been added or removed.
- The system shall process API requests to the cloud-based LLM and generate the initial multi-agent dialogue response within 1.5 seconds, utilizing token streaming to ensure a seamless user experience despite network latency.

2) Resource & Hardware Constraints:

- The cellar's internal computing module shall function strictly as a thin client, capping local memory usage (RAM) at 1 GB and offloading all intensive Vision AI and LLM inferences to the cloud.
- The local computing processes shall be strictly low-power to ensure zero thermal interference with the cellar's precise climate control mechanisms.

3) Reliability & Availability:

- The system shall preserve the submitted data locally so that it remains after a temporary application restart.
- The system shall implement an offline fallback mechanism, retaining basic inventory viewing and temperature control capabilities when the Wi-Fi connection to the cloud is temporarily lost.
- The system shall automatically back up all inventory and configuration data to the cloud servers and support seamless recovery following an unexpected shutdown or hardware replacement.

4) Security & Privacy:

- The system shall require secure authentication (e.g., OAuth 2.0) before a user can access inventory management functions across devices.
- The system shall securely hash and store user passwords on the cloud backend.
- The system shall enforce End-to-End (E2E) encryption (TLS 1.3) for all data transmissions between the cellar appliance, the mobile application, and the LG Datacenter, ensuring personal consumption data and AI interactions cannot be intercepted.
- The system shall strictly isolate user databases on the cloud server to prevent cross-account data leaks.
- The system shall be fully compliant with data protection regulations such as GDPR, offering users the right to export or permanently delete their data from the cloud.

5) Maintainability:

- The system architecture shall support the server-side replacement, scaling, or upgrading of the backend LLMs without requiring firmware updates or modifications to the user interface on the cellar appliance.
- The system architecture shall cleanly separate the frontend UI components from the backend API services to allow independent development and deployment cycles.

6) Usability:

- The system shall provide an intuitive interface allowing new users to navigate core features without consulting external documentation.
- The system shall clearly indicate the current network connection status and authentication state on the main display.
- The system shall preserve unsaved user input locally during sudden network drops, preventing data loss before synchronizing with the cloud.